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ESEIAAT

Computational Engineering

Turbulence: Burgers' equation in Fourier space

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1 Problem Specification

This work concerns the numerical solution of the one-dimensional viscous Burgers equation in a periodic domain using spectral Galerkin truncation. Two solutions are computed: a fully resolved reference solution (DNS) and a Large Eddy Simulation (LES) in which high-wavenumber content is dissipated via a spectral eddy-viscosity model.

The governing equation reads:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}, \qquad x \in [0, 2\pi],\ t > 0, \tag{1}$$

where $u(x,t)$ is the velocity field and ν is the kinematic (molecular) viscosity. We assume periodicity:

$$u(x + 2\pi, t) = u(x, t).$$

On the one hand, Direct Numerical Simulation (DNS) is a numerical approach where all scales of motion are resolved explicitly. In this method, no modeling assumptions are made: the smallest dissipative scales are directly captured by using a sufficiently fine grid or a large number of spectral modes. While DNS provides the most accurate solution, it is computationally very expensive and quickly becomes infeasible for high Reynolds numbers.

On the other hand, Large Eddy Simulation (LES) was desighed with the objective ofreducing computational cost by only resolving the most energy-containing scales of motion (large scale). The smaller scales (which contain lower energy levels) are modeled using an *eddy-viscosity* approach. This introduces an additional viscosity term that mimics the effect of unresolved small-scale turbulence. In this work, the LES formulation uses a spectral eddy-viscosity model that increases dissipation at high wavenumbers. This allows the LES to reproduce the large-scale behavior of the flow with fewer degrees of freedom, at the cost of reduced accuracy in the smallest scales.

2 Theoretical Background

2.1 Fourier Representation

The Fourier representation is fundamental in solving the Burgers equation with spectral methods. Because the equation is defined on a periodic domain, the Fourier basis provides an exact and orthogonal decomposition of the velocity field. This allows us to represent spatial derivatives as simple algebraic operations in spectral space, transforming partial

differential equations into a system of ordinary differential equations for the Fourier coefficients. Moreover, this representation naturally separates contributions by wavenumber, making it easier to study energy transfer between scales.

The velocity field is expanded in Fourier series:

$$u(x,t) = \sum_{k=-N}^{N} \hat{u}_k(t)\, e^{ikx}. \tag{2}$$

In this basis, spatial derivatives are algebraic:

$$\widehat{u_x}_k = ik\hat{u}_k, \qquad \widehat{u_{xx}}_k = -k^2\hat{u}_k.$$

Substituting the expansion into (1) yields the ODE system in spectral space:

$$\frac{d\hat{u}_k}{dt} = -\widehat{uu_x}_k - \nu k^2 \hat{u}_k. \tag{3}$$

2.2 Nonlinear Term as a Convolution

Using the product rule in Fourier space, the nonlinear term becomes a triadic convolution, expressed as:

$$\widehat{uu_x}_k = i \sum_{p=-N}^{N} (k-p)\, \hat{u}_p\, \hat{u}_{k-p}. \tag{4}$$

This transfer term redistributes energy between modes while preserving total energy in the inviscid case. Physically, it represents the nonlinear interaction among different wavenumbers that transfers energy from large to small scales.

2.3 Energy Conservation and Aliasing

For smooth and fully resolved solutions, the nonlinear term is energy-conserving:

$$\sum_{k=-N}^{N} \hat{u}_k^* \widehat{uu_x}_k = 0. \tag{5}$$

However, when the convolution (4) is computed in a truncated basis, the high-frequency modes that are not represented in the system can "fold back" into the resolved range. This effect, known as *aliasing*, occurs because the discrete Fourier transform cannot distinguish between a mode above the Nyquist limit and a lower one that appears identical in the sampled grid. In the context of the Burgers equation, aliasing leads to unphysical energy

transfer between modes, which can artificially increase or decrease total energy.

A common strategy to prevent aliasing is the *de-aliasing* technique, such as the 2/3 rule, where the highest one-third of wavenumbers are set to zero when computing nonlinear products. This removes the spurious energy interactions that would otherwise pollute the lower modes.

3 Numerical Method

3.1 Spectral Galerkin Discretization

We retain modes $k = -N, \ldots, N$ giving $K = 2N + 1$ Fourier coefficients. Each coefficient $\hat{u}_k$ represents the amplitude of a sine or cosine wave of frequency k. The lowest mode, $k = 0$, is known as the *zero mode* and corresponds to the spatial average of the velocity field. Physically, it represents the mean flow component or the net momentum of the system. The higher modes ($|k| > 0$) represent oscillatory components associated with finer spatial structures.

In the implementation, the zero mode is stored at index $\texttt{pk0} = N + 1$ in the code array.

3.2 Time Integration

Time advancement uses explicit Euler:

$$\hat{u}_k^{n+1} = \hat{u}_k^n + \Delta t \left(-\widehat{u u_x}_k^{\,n} - \nu k^2 \hat{u}_k^n \right). \tag{6}$$

The chosen timestep is:

$$\Delta t = \frac{\mathrm{CFL}}{\nu N^2}. \tag{7}$$

The CFL (Courant–Friedrichs–Lewy) condition ensures numerical stability. It constrains the timestep Δt so that information cannot propagate more than one spatial step per iteration. In spectral simulations, the most restrictive time scale is associated with the highest wavenumber, which is why Δt decreases with N^2.

3.3 LES Spectral Eddy-Viscosity Model

The LES simulation augments ν with a wavenumber-dependent eddy-viscosity:

$$\nu_{\text{eff}}(k) = \nu + C_K \left(\frac{|k|}{N}\right)^m, \tag{8}$$

so that the dissipation term becomes

$$D_k^{LES} = -\nu_{\text{eff}}(k) k^2 \hat{u}_k. \tag{9}$$

The purpose of this additional term is to model the dissipation of energy that would normally occur at unresolved scales. Since LES uses fewer modes, the small-scale structures (high-k) cannot be directly simulated. The eddy-viscosity increases with wavenumber, which means that smaller scales experience stronger artificial dissipation. This approach preserves the large-scale dynamics while preventing unphysical energy accumulation at high wavenumbers.

3.4 Forcing

A constant forcing is applied to mode $k = 1$:

$$\hat{u}_1(t) = \text{constant}. \tag{10}$$

This keeps the system dynamically active. The mode $k = 1$ corresponds to the largest possible wavelength in the domain, representing the largest spatial scale. By forcing this mode, we continuously inject energy into the system at large scales, which then cascades to smaller ones through nonlinear interactions. This mimics the physical mechanism of energy input in turbulent flows.

4 Numerical Studies and Physical Interpretation

This section presents and analyzes the results of the simulations, comparing the behavior of the DNS and LES models at different spectral resolutions N.

4.1 Total Energy Evolution

Figure 1 shows the temporal evolution of the total kinetic energy, computed as:

$$E_{\text{tot}}(t) = \sum_{k=-N}^{N} |\hat{u}_k(t)|^2.$$

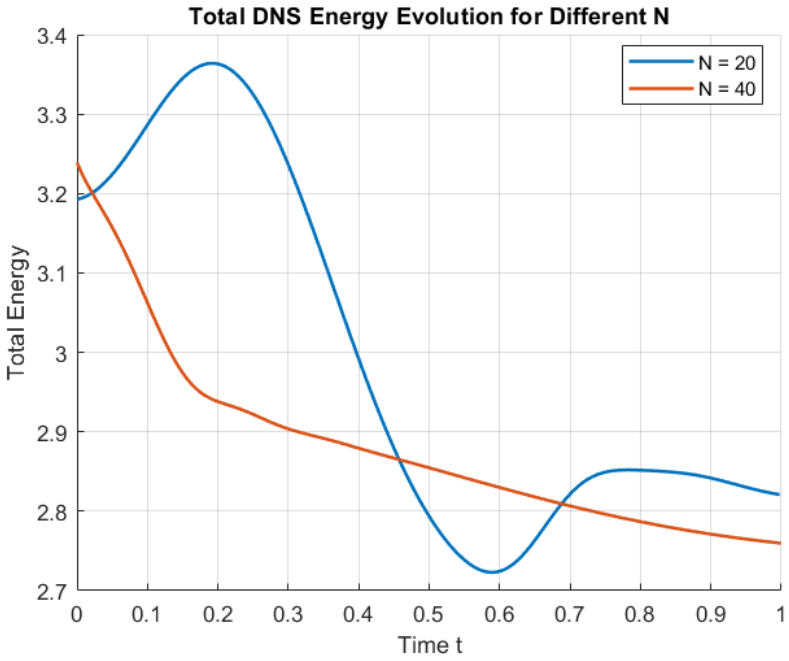


Figure 1: Total DNS energy evolution for different spectral resolutions N.

For both resolutions, the total energy initially increases slightly and then decays over time due to viscous dissipation. The lower-resolution simulation ($N = 20$) exhibits larger oscillations in total energy, which can be attributed to under-resolution and aliasing effects. As the resolution increases to $N = 40$, the solution becomes smoother, and the energy decay is more monotonic, indicating that the higher resolution better captures the true dynamics and dissipative processes.

The overall energy decreases with time as expected, reflecting the dissipative nature of the Burgers equation. The higher-frequency content (small scales) dissipates faster, and the system tends to a steady state. The improved convergence for $N = 40$ suggests that the solution is approaching the true DNS limit.

4.2 Energy Spectrum Comparison

Figure 2 compares the final energy spectra between DNS and LES for two different resolutions.

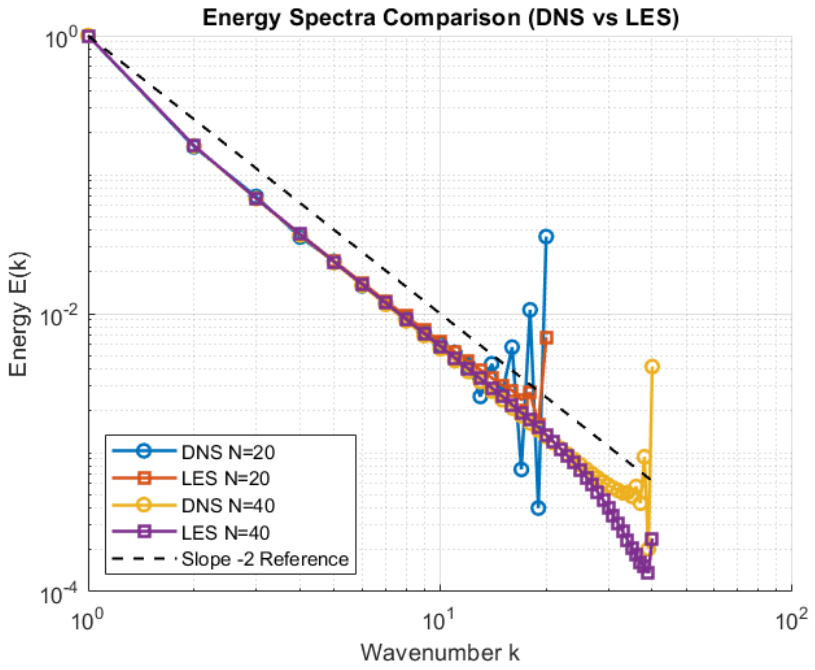


Figure 2: Energy spectra $E(k)$ for DNS and LES at different resolutions.

The DNS spectra follow a clear power-law decay with wavenumber, approaching a k^{-2} trend typical of the 1D Burgers turbulence regime. Increasing the resolution from $N = 20$ to $N = 40$ extends the range of resolved scales and makes the slope closer to the expected theoretical behavior.

The LES spectra show additional damping at high wavenumbers due to the eddy-viscosity model. This indicates that the model successfully removes the excess energy at small scales, preventing aliasing or numerical instabilities. Importantly, the low-wavenumber region (large scales) is nearly identical between DNS and LES, meaning that the LES model preserves the large-scale dynamics while modeling small-scale dissipation effectively.

At the highest resolved wavenumbers, a small pile-up of energy can be observed in DNS for $N = 20$, consistent with aliasing effects discussed earlier. This pile-up disappears in the LES and in the higher-resolution DNS case, confirming that both increased resolution and eddy-viscosity modeling help control spectral contamination.

5 Conclusions

In this project, the one-dimensional viscous Burgers equation was solved using a spectral Galerkin method. Two approaches were compared: a Direct Numerical Simulation (DNS) and a Large Eddy Simulation (LES) with a spectral eddy-viscosity model.

The Fourier representation allowed for an efficient and accurate formulation of the problem in spectral space, transforming derivatives into algebraic terms and enabling a clear view of energy distribution along different wavelengths. Aliasing and under-resolution were shown to affect energy conservation, but their influence diminishes with higher spectral resolution or by applying an eddy-viscosity model.

The results demonstrate that:

- The total energy decays over time due to viscous dissipation, with higher resolution leading to smoother and more physically consistent results.
- The LES model successfully reproduces the large-scale dynamics while damping small-scale fluctuations, preventing numerical energy pile-up at high wavenumbers.
- The energy spectrum approached the expected k^{-2} slope for increasing N, consistent with theoretical predictions for Burgers turbulence (burgulence).

Overall, the LES approach provides a good balance between accuracy and computational efficiency. It captures the essential flow dynamics while reducing the computational cost compared to fully resolved DNS, making it a useful tool for studying turbulence-like phenomena in simplified models.

A Annex: Code

A.1 main.m

```
% Solves the 1D viscous Burgers equation in Fourier space using Galerkin.
% Includes DNS and LES with spectral eddy-viscosity.

clear all; close all; clc;

%% 1 Configuration Parameters
% Read simulation configuration from input function
[N_list, Re, tmax, CFL, CK, m, usr] = readInputData();
nu = 1/Re; % Molecular viscosity

%% 2 Simulation Loop
if usr.doPlots
    figure(1); hold on;
    set(gca, 'XScale', 'log', 'YScale', 'log');
    set(gca, 'XTick', [1 10 100]); set(gca, 'YTick', [1e-6 1e-4 1e-2 1 1e2]);
    grid on;
end

% Preallocating energy and time cells
energy_total_dns = cell(1, length(N_list));
time_all = cell(1, length(N_list));

% Main loop
for idx = 1:length(N_list)
    N = N_list(idx);

    dt = CFL / (nu * N^2); % Time step based on stability condition
    nt = round(tmax/dt);   % Number of time steps

    % Wavenumber vector (Fourier modes)
    k_vec = -N:N;
    K = length(k_vec);
    pk0 = N + 1; % Index of zero mode

    % Initialize spectral solution with |u_k| appx 1/|k|
```

```
uk_dns = zeros(1, K);
for j = 1:N
    uk_dns(pk0 + j) = 1/j;
    uk_dns(pk0 - j) = conj(uk_dns(pk0 + j));
end
uk_dns(pk0) = 0; % Remove mean flow

% LES solution initially matches DNS
uk_les = uk_dns;

% Store large-scale forcing (mode k=1)
uk1_const = uk_dns(pk0 + 1);

% Spectral eddy viscosity model for LES
nu_t = CK * (abs(k_vec) / N).^m;

% Preallocate energy storage
energy_total_dns{idx} = zeros(1, nt);

% Time integration loop
for t = 1:nt
    % Nonlinear term (Galerkin convolution)
    C_dns = compute_convolution(uk_dns, k_vec, K);
    C_les = compute_convolution(uk_les, k_vec, K);

    % Dissipation terms
    D_dns = -(k_vec.^2) * nu .* uk_dns;
    nu_eff_les = nu + nu_t;
    D_les = -(k_vec.^2) .* nu_eff_les .* uk_les;

    % Explicit Euler integration
    uk_dns = uk_dns + dt * (-C_dns + D_dns);
    uk_les = uk_les + dt * (-C_les + D_les);

    % Apply large-scale forcing
    uk_dns(pk0 + 1) = uk1_const;
    uk_dns(pk0 - 1) = conj(uk1_const);
    uk_les(pk0 + 1) = uk1_const;
    uk_les(pk0 - 1) = conj(uk1_const);
```

```
        % Store total DNS energy
        energy_total_dns{idx}(t) = sum(abs(uk_dns).^2);

        % Progress display
        if usr.displayProgress
            if mod(t, round(nt/10)) == 0
                fprintf('N=%d --> %.0f%%\n', N, 100*t/nt);
            end
        end
    end

    % Compute final energy spectra (positive wavenumbers only)
    k_pos = k_vec(pk0+1:end);
    E_k_dns = abs(uk_dns(pk0+1:end)).^2;
    E_k_les = abs(uk_les(pk0+1:end)).^2;

    % Plot energy spectra
    if usr.doPlots
        loglog(k_pos, E_k_dns, 'o-', 'LineWidth', 1.5, 'DisplayName', ['DNS N=' num2st
        loglog(k_pos, E_k_les, 's-', 'LineWidth', 1.5, 'DisplayName', ['LES N=' num2st
    end
    % Store time vector for later plotting
    time_all{idx} = dt * (0:nt-1);
end

if usr.doPlots
    %% 3 Energy Spectrum Plot (Figure 1)
    % Add reference slope for comparison
    k_ref = k_pos(1);
    ref = (k_pos / k_ref).^(-2) * E_k_dns(1);
    loglog(k_pos, ref, '--k', 'LineWidth', 1.2, 'DisplayName', 'Slope -2 Reference');

    xlabel('Wavenumber k');
    ylabel('Energy E(k)');
    title('Energy Spectra Comparison (DNS vs LES)');
    legend('Location', 'southwest');
    hold off;
```

```
    %% 4 Total Energy Evolution (Figure 2)
    figure(2);
    hold on;
    for idx = 1:length(N_list)
        plot(time_all{idx}, energy_total_dns{idx}, 'LineWidth', 1.5, 'DisplayName', [
    end
    xlabel('Time t');
    ylabel('Total Energy');
    title('Total DNS Energy Evolution for Different N');
    grid on;
    legend('Location', 'best');
    hold off;
end
```

A.2 compute_convolution.m

```
function C_k = compute_convolution(uk, k_vec, K)

    C_k = zeros(1, K);
    for j = 1:K

        k = k_vec(j);
        sum_triads = 0;

        for p_idx = 1:K

            p = k_vec(p_idx);
            q = k - p;
            q_idx = find(k_vec == q);

            if ~isempty(q_idx)

                up = uk(p_idx);
                uq = uk(q_idx);
                sum_triads = sum_triads + up * (1i * q) * uq;

            end
        end
        C_k(j) = sum_triads;
```

```
    end
end
```

A.3 readInputData.m

```
function [N_list, Re, tmax, CFL, CK, m, usr] = readInputData()
    % General, numerical and physical
    N_list = [20, 40];
    Re = 40;
    tmax = 1;
    CFL = 0.025;

    % LES
    CK = 0.05;
    m = 2;

    % User Visualization
    usr.displayProgress = true;
    usr.doPlots = true;

end
```